

Chapter 1 & 2:

Introduction to Regression Models and Basic Statistical Concepts

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Conceptual Summary (Core Ideas)

1. Population Regression Function (PRF)

The *theoretical* relationship between the dependent variable Y and the explanatory variable X :

$$E(Y | X) = \beta_1 + \beta_2 X$$

This represents the *expected* or *average value* of Y given a specific value of X .

2. Sample Regression Function (SRF)

Since we don't have access to the entire population, we estimate the parameters β_1, β_2 using sample data:

$$\hat{Y}_i = \hat{\beta}_1 + \hat{\beta}_2 X_i$$

3. Covariance and Correlation

- *Covariance* shows whether X and Y move together (positive or negative).
- *Correlation coefficient* (r) standardizes covariance and lies between -1 and $+1$:

$$r = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

4. Interpretation of $E(Y | X)$

It is the mean (expected value) of Y for a given value of X . As X changes, the expected value of Y also changes.

5. Correlation vs. Causation

Correlation shows association, *not* causality.

Example: Ice cream sales and drowning cases are correlated in summer, but one does not cause the other.

Questions + Answers

1. Why is $E(Y | X)$ called a *conditional mean*?

Because it shows the expected value of Y *given* a specific value of X , not for the population as a whole.

2. What does the correlation coefficient tell us about two variables?

It measures both the *strength* and *direction* of their linear relationship:

$r > 0$ means direct, $r < 0$ inverse, $r = 0$ no linear relation.

3. If $Cov(X, Y) = 0$, does that mean X and Y are independent?

Not necessarily. Independence implies zero covariance, but zero covariance does not always imply independence unless the relationship is linear or both variables are normally distributed.

4. What's the difference between the PRF and the SRF?

The PRF represents the *true*, unobservable relationship in the population, while the SRF is its *empirical estimate* based on sample data.

5. Why do we use the regression function for prediction?

Because the estimated SRF provides the best approximation (in the least-squares sense) of the population's average relationship.

Problem Set (with Answer Guide (Short Solutions))

Problem 2.1 - Conditional Expectation

Suppose the following joint probability distribution between X (years of education) and Y (monthly income in hundreds of dollars) is given:

X (years)	Y (hundreds of \$)	P(X,Y)
10	15	0.1
10	20	0.1
12	25	0.2
12	30	0.2
16	40	0.2
16	45	0.2

Tasks:

1. Compute $E(Y | X = 10)$, $E(Y | X = 12)$, and $E(Y | X = 16)$.
2. Derive the *population regression function (PRF)*: $E(Y | X) = \beta_1 + \beta_2 X$.
3. Interpret β_2 .

Answer Guide 2.1

$$E(Y | X = 10) = (15 \times 0.1 + 20 \times 0.1)/0.2 = 17.5$$

$$E(Y | X = 12) = (25 \times 0.2 + 30 \times 0.2)/0.4 = 27.5$$

$$E(Y | X = 16) = (40 \times 0.2 + 45 \times 0.2)/0.4 = 42.5$$

$$\text{PRF slope: } \beta_2 = (42.5 - 17.5)/(16 - 10) = 25/6 \approx 4.17.$$

Interpretation: each additional year of education increases expected income by \$417.

Problem 2.2 – Covariance and Correlation

You observe the following data:

X	Y
1	2
2	3
3	4
4	6

Tasks:

1. Calculate means $\bar{X}$ and $\bar{Y}$.
2. Compute the covariance $Cov(X, Y)$.
3. Compute the correlation coefficient r .
4. Interpret whether the relationship is strong, weak, positive, or negative.

Answer Guide 2.2

$$\bar{X} = 2.5, \bar{Y} = 3.75.$$

$$Cov(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n-1} = 1.67.$$

$$r = 1.67 / (1.29 \times 1.71) = 0.75.$$

→ moderately strong positive linear relationship.

Problem 2.3 – Interpreting Correlation vs. Causality

A researcher finds that ice cream sales and drowning incidents have $r = 0.85$.

Tasks:

1. Explain why this does not imply causality.
2. Suggest what hidden variable might explain this correlation.

Answer Guide 2.3

Correlation $\neq$ causation; both are driven by temperature (a confounding variable).